Полный отчет анализа моделей

Сводные результаты

Серия	Модель	Средняя MAE	Средняя RMSE	Средняя sMAPE	Среднее время обучения	Лучшая модель
A	LSTM (128u, 12w, 100e, 16b)	0.1228	0.1390	4.1402%	18.92 сек	
A	LSTM (128u, 12w, 100e, 32b)	0.1078	0.1286	3.7179%	16.38 сек	
A	LSTM (128u, 12w, 100e, 64b)	0.2212	0.2397	7.7486%	14.53 сек	
A	LSTM (128u, 12w, 150e, 16b)	0.4239	0.4584	15.4755%	25.52 сек	
A	LSTM (128u, 12w, 150e, 32b)	0.2310	0.2500	8.1059%	20.99 сек	
A	LSTM (128u, 12w, 150e, 64b)	0.1835	0.2005	6.3843%	18.74 сек	
A	LSTM (128u, 12w, 50e, 16b)	0.3895	0.4216	14.1162%	12.90 сек	
A	LSTM (128u, 12w, 50e, 32b)	0.2335	0.2525	8.1963%	11.67 сек	
A	LSTM (128u, 12w, 50e, 64b)	0.1882	0.2052	6.5554%	12.16 сек	
A	LSTM (128u, 18w, 100e, 16b)	0.0660	0.0941	2.2860%	22.24 сек	
A	LSTM (128u, 18w, 100e, 32b)	0.0986	0.1203	3.3978%	19.86 сек	
A	LSTM (128u, 18w, 100e, 64b)	0.2020	0.2194	7.0497%	17.28 сек	
A	LSTM (128u, 18w, 150e, 16b)	0.3749	0.4052	13.5437%	30.45 сек	
A	LSTM (128u, 18w, 150e, 32b)	0.1015	0.1228	3.4968%	24.64 сек	
A	LSTM (128u, 18w, 150e, 64b)	0.1903	0.2076	6.6291%	21.69 сек	
A	LSTM (128u, 18w, 50e, 16b)	0.3347	0.3622	11.9890%	14.87 сек	
A	LSTM (128u, 18w, 50e, 32b)	0.0671	0.0942	2.3220%	13.39 сек	
A	LSTM (128u, 18w, 50e, 64b)	0.1804	0.1975	6.2756%	28.19 сек	
A	LSTM (128u, 24w, 100e, 16b)	0.2802	0.3028	9.9247%	24.36 сек	
A	LSTM (128u, 24w, 100e, 32b)	0.2268	0.2459	7.9532%	20.55 сек	
A	LSTM (128u, 24w, 100e, 64b)	0.1948	0.2123	6.7920%	18.70 сек	
A	LSTM (128u, 24w, 150e, 16b)	0.2044	0.2227	7.1347%	33.79 сек	
A	LSTM (128u, 24w, 150e, 32b)	0.0634	0.0945	2.1982%	27.14 сек	+
A	LSTM (128u, 24w, 150e, 64b)	0.2240	0.2427	7.8491%	23.51 сек	
A	LSTM (128u, 24w, 50e, 16b)	0.2222	0.2410	7.7852%	16.32 сек	
A	LSTM (128u, 24w, 50e, 32b)	0.1743	0.1916	6.0551%	14.38 сек	
A	LSTM (128u, 24w, 50e, 64b)	0.1817	0.1985	6.3221%	13.07 сек	
A	LSTM (128u, 3w, 100e, 16b)	0.1838	0.2035	6.3897%	15.57 сек	
A	LSTM (128u, 3w, 100e, 32b)	0.1528	0.1715	5.2891%	13.12 сек	

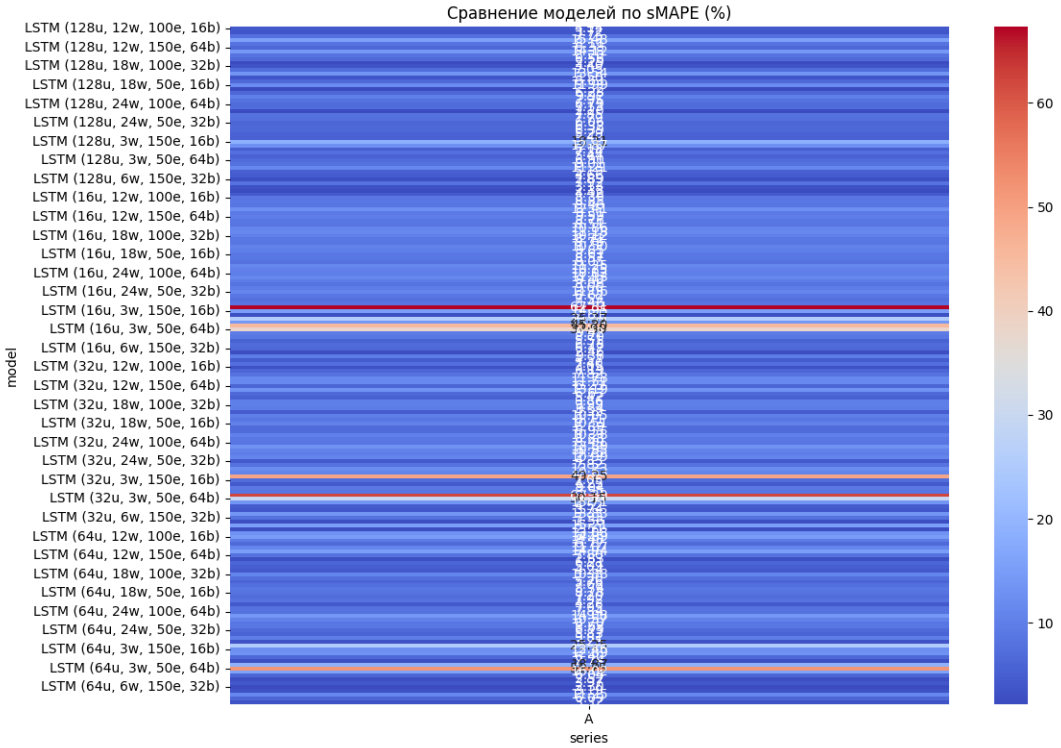
A	LSTM (128u, 3w, 100e, 64b)	0.1646	0.1992	5.4010%	12.35 сек	
A	LSTM (128u, 3w, 150e, 16b)	0.4932	0.5406	18.3135%	17.48 сек	
A	LSTM (128u, 3w, 150e, 32b)	0.3572	0.3932	12.8654%	15.49 сек	
A	LSTM (128u, 3w, 150e, 64b)	0.1496	0.1676	5.1797%	14.79 сек	
A	LSTM (128u, 3w, 50e, 16b)	0.2102	0.2305	7.3447%	19.47 сек	
A	LSTM (128u, 3w, 50e, 32b)	0.1657	0.1989	5.4384%	10.43 сек	
A	LSTM (128u, 3w, 50e, 64b)	0.2147	0.2655	6.9028%	10.01 сек	
A	LSTM (128u, 6w, 100e, 16b)	0.2284	0.2495	8.0082%	15.61 сек	
A	LSTM (128u, 6w, 100e, 32b)	0.3142	0.3438	11.2101%	13.82 сек	
A	LSTM (128u, 6w, 100e, 64b)	0.1500	0.1681	5.1912%	12.95 сек	
A	LSTM (128u, 6w, 150e, 16b)	0.1390	0.1558	4.6552%	19.71 сек	
A	LSTM (128u, 6w, 150e, 32b)	0.0844	0.1050	2.8944%	17.15 сек	
A	LSTM (128u, 6w, 150e, 64b)	0.2259	0.2466	7.9155%	15.87 сек	
A	LSTM (128u, 6w, 50e, 16b)	0.0916	0.1104	3.1307%	11.29 сек	
A	LSTM (128u, 6w, 50e, 32b)	0.0685	0.0974	2.3782%	10.53 сек	
A	LSTM (128u, 6w, 50e, 64b)	0.1280	0.1474	4.4215%	9.96 сек	
A	LSTM (16u, 12w, 100e, 16b)	0.2388	0.2588	8.3891%	17.23 сек	
A	LSTM (16u, 12w, 100e, 32b)	0.2514	0.2718	8.8542%	14.78 сек	
A	LSTM (16u, 12w, 100e, 64b)	0.2408	0.2607	8.4635%	13.32 сек	
A	LSTM (16u, 12w, 150e, 16b)	0.3455	0.3724	12.4058%	21.56 сек	
A	LSTM (16u, 12w, 150e, 32b)	0.2636	0.2851	9.3093%	18.00 сек	
A	LSTM (16u, 12w, 150e, 64b)	0.2713	0.2934	9.5944%	16.08 сек	
A	LSTM (16u, 12w, 50e, 16b)	0.2488	0.2690	8.7579%	12.62 сек	
A	LSTM (16u, 12w, 50e, 32b)	0.2421	0.2623	8.5141%	11.08 сек	
A	LSTM (16u, 12w, 50e, 64b)	0.3050	0.3296	10.8610%	11.22 сек	
A	LSTM (16u, 18w, 100e, 16b)	0.3134	0.3380	11.1788%	19.45 сек	
A	LSTM (16u, 18w, 100e, 32b)	0.2853	0.3081	10.1196%	16.93 сек	
A	LSTM (16u, 18w, 100e, 64b)	0.2474	0.2679	8.7093%	15.29 сек	
A	LSTM (16u, 18w, 150e, 16b)	0.2488	0.2693	8.7596%	23.85 сек	
A	LSTM (16u, 18w, 150e, 32b)	0.2953	0.3188	10.4951%	18.38 сек	
A	LSTM (16u, 18w, 150e, 64b)	0.2781	0.3005	9.8476%	16.07 сек	
A	LSTM (16u, 18w, 50e, 16b)	0.2448	0.2657	8.6096%	13.07 сек	
A	LSTM (16u, 18w, 50e, 32b)	0.2437	0.2640	8.5703%	12.61 сек	
A	LSTM (16u, 18w, 50e, 64b)	0.2294	0.2486	8.0453%	10.80 сек	
A	LSTM (16u, 24w, 100e, 16b)	0.3152	0.3395	11.2452%	17.84 сек	

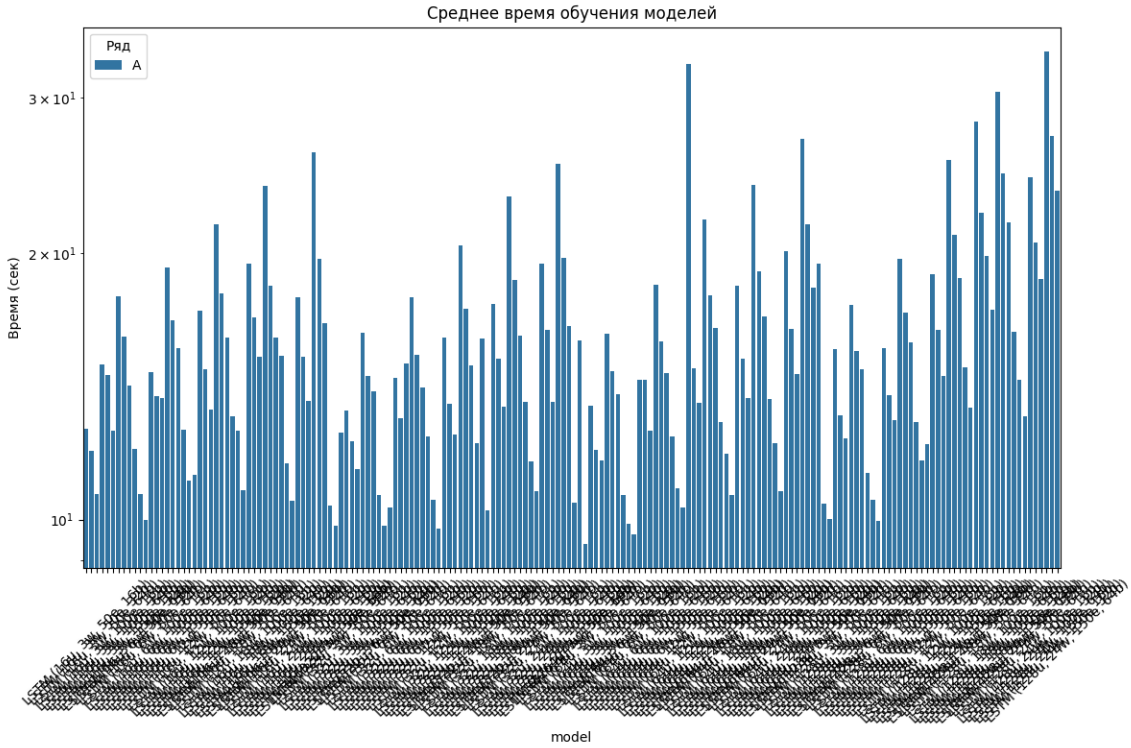
A	LSTM (16u, 24w, 100e, 32b)	0.2890	0.3119	10.2550%	15.27 сек	
A	LSTM (16u, 24w, 100e, 64b)	0.2826	0.3052	10.0170%	13.62 сек	
A	LSTM (16u, 24w, 150e, 16b)	0.3121	0.3368	11.1302%	25.98 сек	
A	LSTM (16u, 24w, 150e, 32b)	0.2661	0.2874	9.4005%	19.73 сек	
A	LSTM (16u, 24w, 150e, 64b)	0.2296	0.2491	8.0552%	16.69 сек	
A	LSTM (16u, 24w, 50e, 16b)	0.2731	0.2949	9.6599%	15.31 сек	
A	LSTM (16u, 24w, 50e, 32b)	0.3102	0.3344	11.0559%	11.58 сек	
A	LSTM (16u, 24w, 50e, 64b)	0.2450	0.2650	8.6214%	10.50 сек	
A	LSTM (16u, 3w, 100e, 16b)	0.2170	0.2386	7.5895%	14.99 сек	
A	LSTM (16u, 3w, 100e, 32b)	0.2649	0.3104	8.4648%	14.55 сек	
A	LSTM (16u, 3w, 100e, 64b)	3.9448	4.9438	67.3105%	12.61 сек	
A	LSTM (16u, 3w, 150e, 16b)	0.4103	0.4484	14.9499%	17.89 сек	
A	LSTM (16u, 3w, 150e, 32b)	0.0812	0.1085	2.8051%	16.09 сек	
A	LSTM (16u, 3w, 150e, 64b)	0.9277	1.1394	25.3659%	14.16 сек	
A	LSTM (16u, 3w, 50e, 16b)	0.4782	0.6096	14.2433%	12.67 сек	
A	LSTM (16u, 3w, 50e, 32b)	2.0622	2.6362	45.1978%	11.96 сек	
A	LSTM (16u, 3w, 50e, 64b)	0.9312	0.9737	37.8993%	10.69 сек	
A	LSTM (16u, 6w, 100e, 16b)	0.2667	0.2912	9.4213%	14.70 сек	
A	LSTM (16u, 6w, 100e, 32b)	0.2474	0.2704	8.7047%	13.79 сек	
A	LSTM (16u, 6w, 100e, 64b)	0.1670	0.1926	5.5083%	13.70 сек	
A	LSTM (16u, 6w, 150e, 16b)	0.1948	0.2160	6.7838%	19.27 сек	
A	LSTM (16u, 6w, 150e, 32b)	0.1860	0.2059	6.4688%	16.81 сек	
A	LSTM (16u, 6w, 150e, 64b)	0.0713	0.0977	2.4627%	15.63 сек	
A	LSTM (16u, 6w, 50e, 16b)	0.2650	0.2893	9.3591%	12.02 сек	
A	LSTM (16u, 6w, 50e, 32b)	0.1323	0.1523	4.5652%	10.68 сек	
A	LSTM (16u, 6w, 50e, 64b)	0.2116	0.2314	7.3962%	9.99 сек	
A	LSTM (32u, 12w, 100e, 16b)	0.0891	0.1070	3.0511%	16.06 сек	
A	LSTM (32u, 12w, 100e, 32b)	0.1780	0.1954	6.1875%	13.53 сек	
A	LSTM (32u, 12w, 100e, 64b)	0.2232	0.2417	7.8206%	12.48 сек	
A	LSTM (32u, 12w, 150e, 16b)	0.3293	0.3561	11.7844%	20.42 сек	
A	LSTM (32u, 12w, 150e, 32b)	0.3222	0.3484	11.5125%	17.32 сек	
A	LSTM (32u, 12w, 150e, 64b)	0.1791	0.1962	6.2288%	14.92 сек	
A	LSTM (32u, 12w, 50e, 16b)	0.3658	0.3955	13.1891%	12.41 сек	
A	LSTM (32u, 12w, 50e, 32b)	0.1903	0.2080	6.6265%	10.53 сек	
A	LSTM (32u, 12w, 50e, 64b)	0.1564	0.1735	5.4183%	9.77 сек	

A	LSTM (32u, 18w, 100e, 16b)	0.2785	0.3007	9.8636%	17.52 сек	
A	LSTM (32u, 18w, 100e, 32b)	0.2820	0.3045	9.9938%	15.20 сек	
A	LSTM (32u, 18w, 100e, 64b)	0.2697	0.2916	9.5361%	13.40 сек	
A	LSTM (32u, 18w, 150e, 16b)	0.1312	0.1514	4.5225%	23.19 сек	
A	LSTM (32u, 18w, 150e, 32b)	0.2836	0.3062	10.0542%	18.64 сек	
A	LSTM (32u, 18w, 150e, 64b)	0.2475	0.2677	8.7127%	16.13 сек	
A	LSTM (32u, 18w, 50e, 16b)	0.2956	0.3190	10.5052%	12.21 сек	
A	LSTM (32u, 18w, 50e, 32b)	0.1976	0.2157	6.8904%	16.01 сек	
A	LSTM (32u, 18w, 50e, 64b)	0.1993	0.2170	6.9516%	10.25 сек	
A	LSTM (32u, 24w, 100e, 16b)	0.2896	0.3124	10.2780%	19.47 сек	
A	LSTM (32u, 24w, 100e, 32b)	0.2521	0.2726	8.8820%	16.37 сек	
A	LSTM (32u, 24w, 100e, 64b)	0.2414	0.2610	8.4856%	13.57 сек	
A	LSTM (32u, 24w, 150e, 16b)	0.3582	0.3855	12.8930%	25.25 сек	
A	LSTM (32u, 24w, 150e, 32b)	0.2914	0.3144	10.3461%	19.78 сек	
A	LSTM (32u, 24w, 150e, 64b)	0.2755	0.2975	9.7495%	16.55 сек	
A	LSTM (32u, 24w, 50e, 16b)	0.3085	0.3328	10.9940%	13.57 сек	
A	LSTM (32u, 24w, 50e, 32b)	0.1256	0.1453	4.3323%	11.63 сек	
A	LSTM (32u, 24w, 50e, 64b)	0.2233	0.2422	7.8240%	10.78 сек	
A	LSTM (32u, 3w, 100e, 16b)	0.3407	0.3747	12.2286%	13.27 сек	
A	LSTM (32u, 3w, 100e, 32b)	0.3423	0.4204	10.6318%	12.25 сек	
A	LSTM (32u, 3w, 100e, 64b)	2.4679	3.3131	49.2464%	11.41 сек	
A	LSTM (32u, 3w, 150e, 16b)	0.2022	0.2232	7.0525%	16.27 сек	
A	LSTM (32u, 3w, 150e, 32b)	0.1232	0.1442	4.2491%	14.52 сек	
A	LSTM (32u, 3w, 150e, 64b)	0.2867	0.3507	9.0422%	13.96 сек	
A	LSTM (32u, 3w, 50e, 16b)	0.2696	0.3279	8.5502%	10.37 сек	
A	LSTM (32u, 3w, 50e, 32b)	3.9785	5.5614	62.3932%	9.84 сек	
A	LSTM (32u, 3w, 50e, 64b)	1.2581	1.7978	30.1520%	12.52 сек	
A	LSTM (32u, 6w, 100e, 16b)	0.3789	0.4138	13.7084%	14.45 сек	
A	LSTM (32u, 6w, 100e, 32b)	0.1308	0.1506	4.5170%	13.03 сек	
A	LSTM (32u, 6w, 100e, 64b)	0.1110	0.1321	3.7414%	15.01 сек	
A	LSTM (32u, 6w, 150e, 16b)	0.3819	0.4166	13.8257%	17.81 сек	
A	LSTM (32u, 6w, 150e, 32b)	0.2262	0.2474	7.9278%	15.36 сек	
A	LSTM (32u, 6w, 150e, 64b)	0.0737	0.1041	2.5588%	14.09 сек	
A	LSTM (32u, 6w, 50e, 16b)	0.4169	0.4569	15.2106%	10.66 сек	
A	LSTM (32u, 6w, 50e, 32b)	0.0726	0.0981	2.5086%	9.85 сек	

A	LSTM (32u, 6w, 50e, 64b)	0.4310	0.5332	13.0780%	10.33 сек	
A	LSTM (64u, 12w, 100e, 16b)	0.4091	0.4417	14.8878%	32.68 сек	
A	LSTM (64u, 12w, 100e, 32b)	0.2681	0.2897	9.4759%	14.82 сек	
A	LSTM (64u, 12w, 100e, 64b)	0.1942	0.2113	6.7695%	13.56 сек	
A	LSTM (64u, 12w, 150e, 16b)	0.3107	0.3360	11.0749%	21.84 сек	
A	LSTM (64u, 12w, 150e, 32b)	0.4104	0.4434	14.9406%	17.93 сек	
A	LSTM (64u, 12w, 150e, 64b)	0.2225	0.2409	7.7953%	16.48 сек	
A	LSTM (64u, 12w, 50e, 16b)	0.0826	0.1023	2.8344%	12.40 сек	
A	LSTM (64u, 12w, 50e, 32b)	0.1785	0.1949	6.2070%	10.86 сек	
A	LSTM (64u, 12w, 50e, 64b)	0.1471	0.1644	5.0928%	10.32 сек	
A	LSTM (64u, 18w, 100e, 16b)	0.1229	0.1428	4.2365%	18.37 сек	
A	LSTM (64u, 18w, 100e, 32b)	0.3055	0.3300	10.8793%	15.22 сек	
A	LSTM (64u, 18w, 100e, 64b)	0.2089	0.2269	7.3007%	13.72 сек	
A	LSTM (64u, 18w, 150e, 16b)	0.1661	0.1839	5.7624%	23.91 сек	
A	LSTM (64u, 18w, 150e, 32b)	0.2192	0.2378	7.6758%	19.09 сек	
A	LSTM (64u, 18w, 150e, 64b)	0.2482	0.2682	8.7395%	16.95 сек	
A	LSTM (64u, 18w, 50e, 16b)	0.2357	0.2551	8.2788%	12.89 сек	
A	LSTM (64u, 18w, 50e, 32b)	0.2061	0.2237	7.2004%	11.88 сек	
A	LSTM (64u, 18w, 50e, 64b)	0.2259	0.2446	7.9200%	10.66 сек	
A	LSTM (64u, 24w, 100e, 16b)	0.1242	0.1444	4.2814%	20.09 сек	
A	LSTM (64u, 24w, 100e, 32b)	0.2267	0.2456	7.9490%	16.43 сек	
A	LSTM (64u, 24w, 100e, 64b)	0.2236	0.2424	7.8362%	14.59 сек	
A	LSTM (64u, 24w, 150e, 16b)	0.3887	0.4183	14.0834%	26.90 сек	
A	LSTM (64u, 24w, 150e, 32b)	0.2973	0.3207	10.5685%	21.58 сек	
A	LSTM (64u, 24w, 150e, 64b)	0.2472	0.2672	8.7010%	18.28 сек	
A	LSTM (64u, 24w, 50e, 16b)	0.2453	0.2651	8.6296%	13.68 сек	
A	LSTM (64u, 24w, 50e, 32b)	0.1937	0.2115	6.7519%	12.19 сек	
A	LSTM (64u, 24w, 50e, 64b)	0.1678	0.1847	5.8255%	10.77 сек	
A	LSTM (64u, 3w, 100e, 16b)	0.2770	0.3036	9.8081%	13.45 сек	
A	LSTM (64u, 3w, 100e, 32b)	0.0939	0.1185	3.2481%	11.99 сек	
A	LSTM (64u, 3w, 100e, 64b)	0.9670	1.3152	25.2542%	11.66 сек	
A	LSTM (64u, 3w, 150e, 16b)	0.3467	0.3811	12.4604%	16.22 сек	
A	LSTM (64u, 3w, 150e, 32b)	0.3577	0.3938	12.8876%	14.72 сек	
A	LSTM (64u, 3w, 150e, 64b)	0.1973	0.2392	6.3980%	13.85 сек	
A	LSTM (64u, 3w, 50e, 16b)	0.0798	0.1054	2.7399%	10.46 сек	

A	LSTM (64u, 3w, 50e, 32b)	0.6694	0.8852	18.8731%	15.93 сек	
A	LSTM (64u, 3w, 50e, 64b)	3.1189	4.7664	51.7503%	9.39 сек	
A	LSTM (64u, 6w, 100e, 16b)	0.4371	0.4779	16.0184%	14.38 сек	
A	LSTM (64u, 6w, 100e, 32b)	0.1739	0.1923	6.0391%	14.37 сек	
A	LSTM (64u, 6w, 100e, 64b)	0.0727	0.0998	2.5219%	12.60 сек	
A	LSTM (64u, 6w, 150e, 16b)	0.1175	0.1337	3.9731%	18.44 сек	
A	LSTM (64u, 6w, 150e, 32b)	0.0663	0.0988	2.3032%	15.89 сек	
A	LSTM (64u, 6w, 150e, 64b)	0.0922	0.1163	3.1859%	14.64 сек	
A	LSTM (64u, 6w, 50e, 16b)	0.3153	0.3449	11.2519%	10.67 сек	
A	LSTM (64u, 6w, 50e, 32b)	0.1754	0.1928	6.0936%	9.89 сек	
A	LSTM (64u, 6w, 50e, 64b)	0.0979	0.1220	3.3163%	9.63 сек	




Среднее время обучения моделей
Ряд
A
3×10^1
2×10^1
10^1
Время (сек)
model

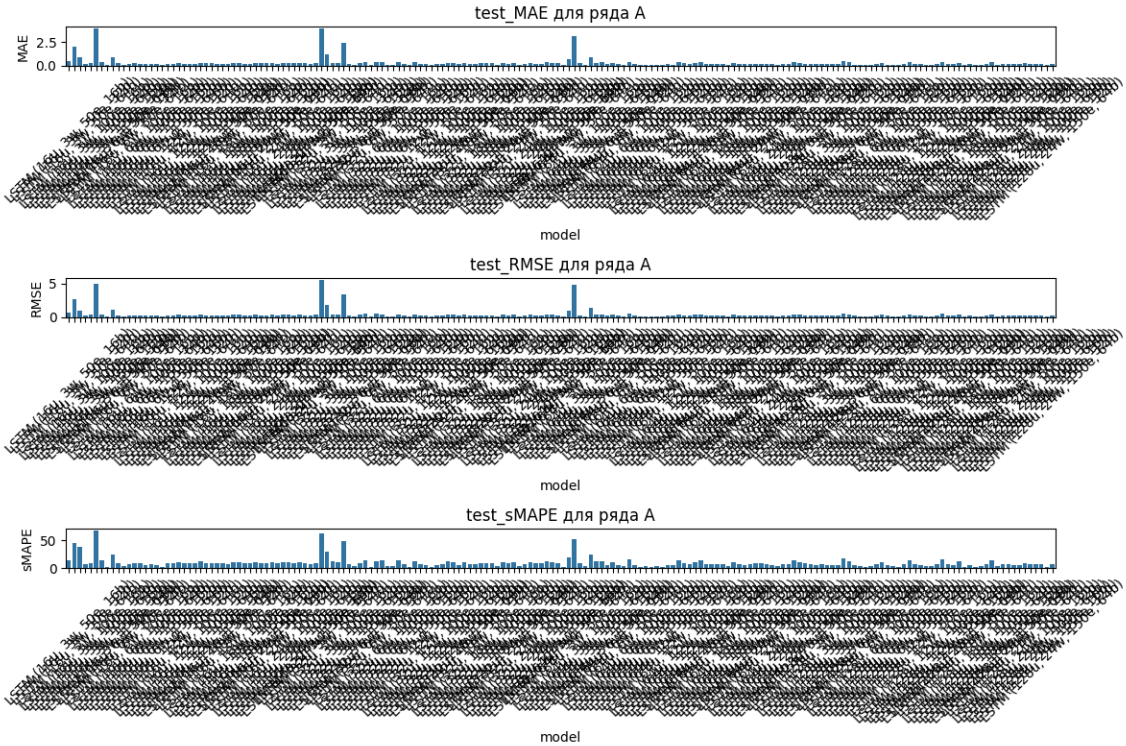
Анализ ряда A

Лучшая модель:

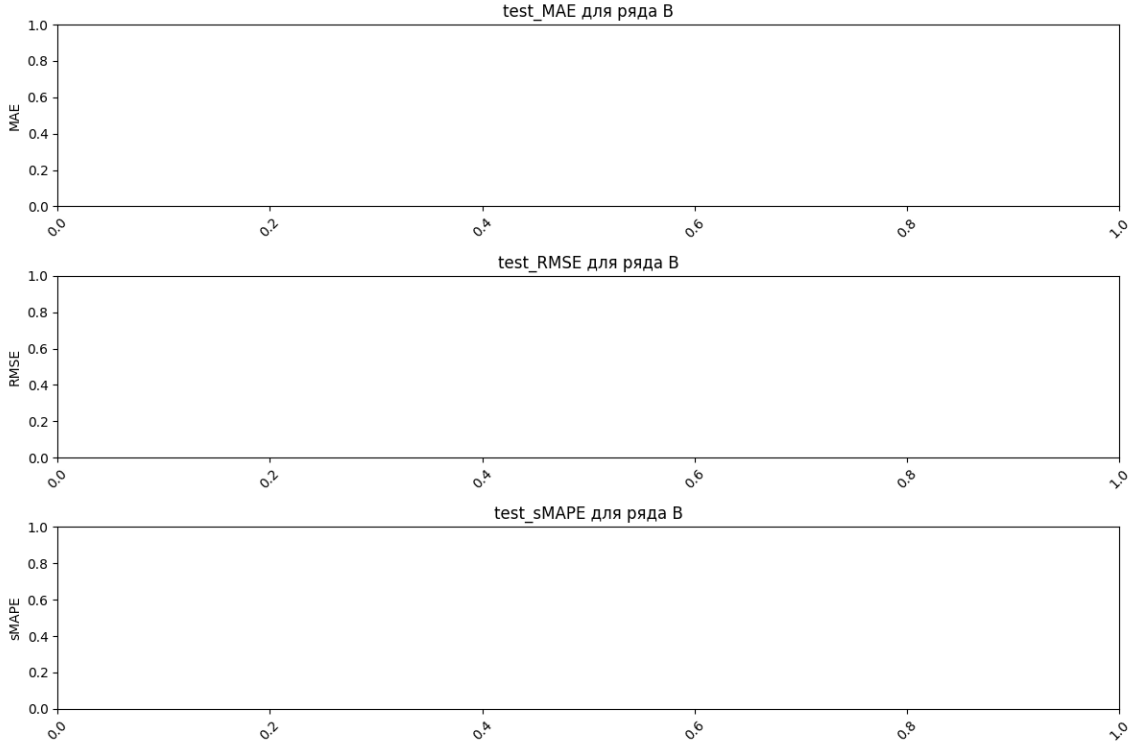
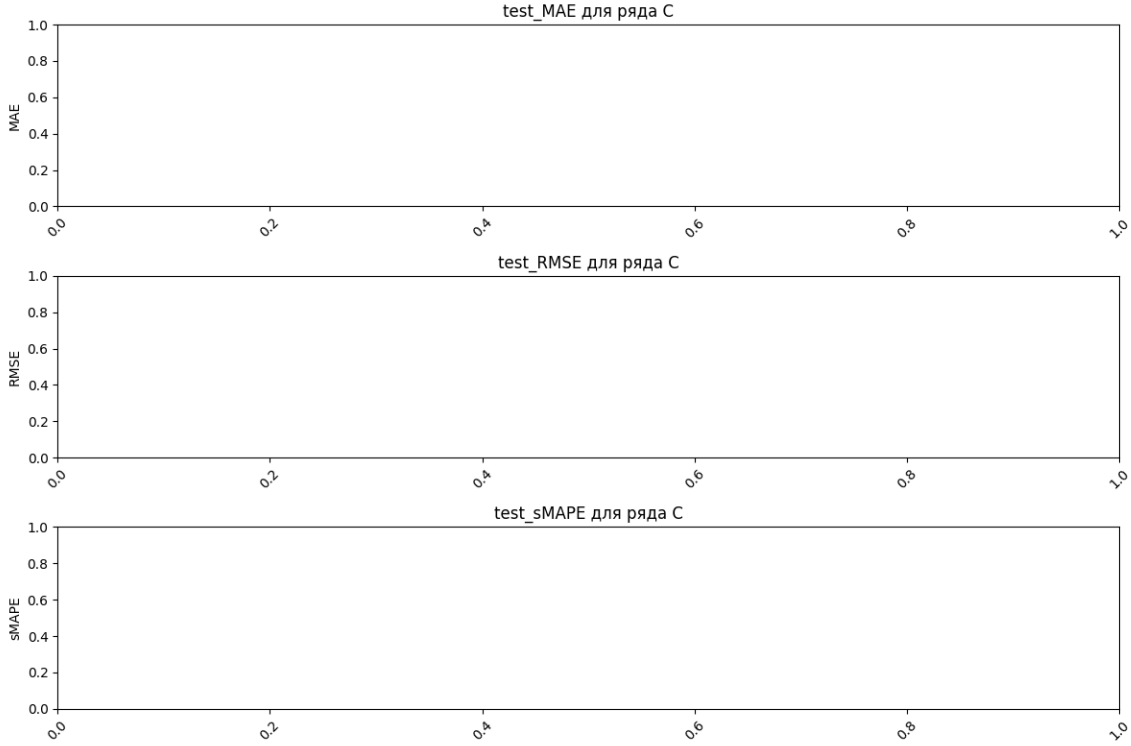
178
A
LSTM (128u, 24w, 150e, 32b)
27.136617183685303
0.2141354501697395
0.22807414743859805
7.621184856946474
0.06339502206421961
0.09448016974262295
2.1981502410189218
False

Детали модели:

Лучшая модель для ряда A: LSTM (128u, 24w, 150e, 32b)
Тестовые метрики:
MAE: 0.0634
RMSE: 0.0945
sMAPE: 2.1982%
Время обучения: 27.14 сек


test_MAE для ряда A
MAE
2.5
0.0
model
test_RMSE для ряда A
RMSE
5
0
model
test_sMAPE для ряда A
sMAPE
50
0
model

Анализ ряда B



Анализ ряда C